

100 DAYS OF TQM : CASE STUDY DAY 27/100

Global Supplier Quality and Vendor Performance Management Strategy

Integrated AS9115 Audits, Variation Tracking, and Receiving Poka Yoke Framework

1. EXECUTIVE SUMMARY AND CONTEXT

Airbus operates one of the most complex cross border aerospace supply chain in the world, receiving critical components from hundreds of Tier 1 vendors across France, Germany, Spain, and UK. Ensuring that every incoming sub assembly matches exact design specifications is vital to prevent structural assembly disruptions. This directive outlines a standardized quality framework using AS9115 audits, dynamic supplier scorecarding, Six Sigma tolerance variation tracking, and material receiving Poka Yoke to ensure zero non conforming parts reach the final assembly line.

2. OPERATIONAL PROBLEM AND ASSEMBLY RISKS

Minor dimensional variations or specification drift in supplied parts relative to engineering blueprints can cause severe bottlenecks during structural integration.

- Specification Drift:** Inconsistency in supplier manufacturing processes leading to parts that deviate slightly from official Airbus engineering drawings.
- Delayed Corrective Actions:** Slow resolution of Supplier Corrective Action Requests (SCAR) creating recurring defect patterns across batches.
- Receiving Errors:** Human mistakes at the receiving bay leading to incorrect part orientation or mix ups during lot intake.

3. TQM ARCHITECTURE AND QUALITY CONTROL TOOLS

- AS9115 and ISO 19011 Supplier Audits:** Standardized on site auditing protocols to evaluate vendor tool calibration, process capability, and quality management readiness before dispatch.
- 4 Pillar Dynamic Supplier Scorecard:** Continuous vendor evaluation tracking Quality (defect rates), Delivery (OTIF), Audit Compliance, and SCAR responsiveness.
- Six Sigma and Cpk Variation Control:** Used specifically to measure dimensional variations in incoming parts against Airbus design blueprints to verify if supplier outputs remain strictly within specified tolerance limits.
- Assembly to Vendor Closed Loop Feedback:** Real time shopfloor feedback routing component assembly issues back to vendor SCAR logs to trigger immediate 8D root cause investigations.

Receiving Bay Poka Yoke (Mistake Proofing Integration):

Automated barcode scanning connected with digital blueprint specs prevents wrong material logging, while physical keying mechanisms at receiving jigs stop misaligned or out of spec components from entering storage.

4. PROCESS CONTROL AND IMPLEMENTATION MATRIX

Operational Stage	Applied Quality Tool	Primary Quality Objective
Vendor Manufacturing	AS9115 / ISO 19011 Audits	Verify process capability and tool calibration standards.
Material Receiving	Receiving Poka Yoke Jigs	Prevent intake of incorrect specs or misoriented components.
Incoming QA	Six Sigma and Cpk Tracking	Monitor dimensional variation against official engineering drawings.
Shop Floor Integration	Closed Loop SCAR System	Automate defect feedback and enforce vendor 8D corrective actions.

Disclaimer: This case study is developed strictly for educational purposes as part of the 100 Days Of TQM learning challenge. The author is not affiliated with, endorsed by, or officially connected to Airbus. This analysis addresses potential operational scenarios and quality engineering frameworks from an academic perspective.